

# Key Terms

**Kilowatt (kW):** Unit of electric power

**Kilowatt-hour (kWh):** Energy unit. One unit is defined as the energy consumed by the power consumption of 1kW during 1-hour.

**Price of electricity:** Cents per kilowatt-hour

**Utility Company:** The organization which provides services to the general public, although it may be privately owned. Public utilities include electric, gas, telephone, water, and television cable systems. They are allowed certain monopoly rights due to the practice need to service entire geographic areas with one system, but they are regulated by the state, county, and/ or city public utility commissions under state law.

**PV:** Photovoltaic cell is a specialized semiconductor that converts visible light into direct current. (DC)

**Inverter:** The inverter device converts the DC electricity generated by the solar panels into alternating current (AC) electricity.

**Array:** Set of solar panels within a section

**Net Metering:** It's a billing mechanism that credits homeowners for the solar electricity they add to the grid. For example, when a homeowner produces more energy with their solar system, the excess power gets exported back into the grid and helps with neighboring customer loads. The homeowner receives energy credits for exported energy into the grid that counts against any energy consumed by the homeowner at night or other periods where the home's electricity use exceeds the output of the system. Homeowners are only billed for their "net" energy use.

**Interconnection:** This is the process in which a homeowner will need to apply for in order for their solar system can get connected with their utility provider. The utility provider will get notified that you are looking into installing a PV Solar system and you are applying to get NET METERING for your excess production. Some utilities provide Solar-Buy Backs which homeowners agree to get a set price per kWh for their exported electricity. Please do your research for your customer's Utility company. As each utility have their own interconnection policies and requirements.

**ITC:** Investment Tax Credit, it's a Federal Tax incentive for homeowners that invest in renewable sources to reduce their carbon emissions.

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**Annual Degradation:** The Solar panels we use at Initiate Solar are considered Tier 1 panels. Meaning high production ratings and low production loss over the years. Solar Panels will lose about 0.05% of their production capabilities every year. In 30 years, consumers will lose 15% energy production loss.

**Solar Panel Efficiency:** Two main factors have changed over time regarding photovoltaic technology, design, and efficiency. The solar panel design has become more sleek, modern, with a black-on-black finish, making the aesthetic appealing. The efficiency has gotten better over the years but it does not work like other technologies like our phones. Panel Efficiency 10 years ago peaked at around 15-16%, and today the average Tier 1 panel produces 19-21% efficiency

**PPA:** A Solar Power Purchase Agreement where a third party owns, operates, and maintains the PV Solar System on their rooftop or property and purchases the power produced by the solar system for the length of the agreement. The homeowner only pays for the actual power produced by the solar system.

**Lease:** Similar to PPA but with a different payment structure. With a lease, you pay the first year's estimated production and divide the cost per kWh in 12 months. Usually comes with a production guarantee.

**Power Production Guarantee:** The guaranteed amount of kWh the system is going to be produced throughout the lifetime of the agreement.

**Tax Incentive:** A tax incentive is an aspect of a country's tax code designed to incentivize, or encourage a particular economic activity.

**Rebate:** A government incentive that all customers would receive regardless of tax appetite

**Shade Report:** Analysis of what times of the day shadows are cast on the roof from trees or other structures that will determine the output of panels.

**PPW:** Price Per Watt

**EPC:** Engineering Procurement Construction

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**Site Audit:** The inspection of the home, site audit team checks, the main electrical panel, meter, attic, rooftop to take account for roof conditions or if there's any obstruction that was not seen on the Original proposal. The site audit team also conducts a shade study to take account of any trees or obstructions that may cause loss of production.

**Final Layout:** Final CAD drawing for a solar project, if there are no changes to the agreement the project moves forward to permitting. If there are needed adders for the project to be doable for example, the home needs a Main panel upgrade, re-roof, upsizing, or downsizing the solar system a change order will be issued With the final scope of work for the homeowner To approve and proceed with a solar project.

**Micro-Inverter:** A small device placed on the back of each individual solar panel that absorbs and converts DC electricity to AC electricity for the use of the home.  
(We use Enphase Micro-Inverters)

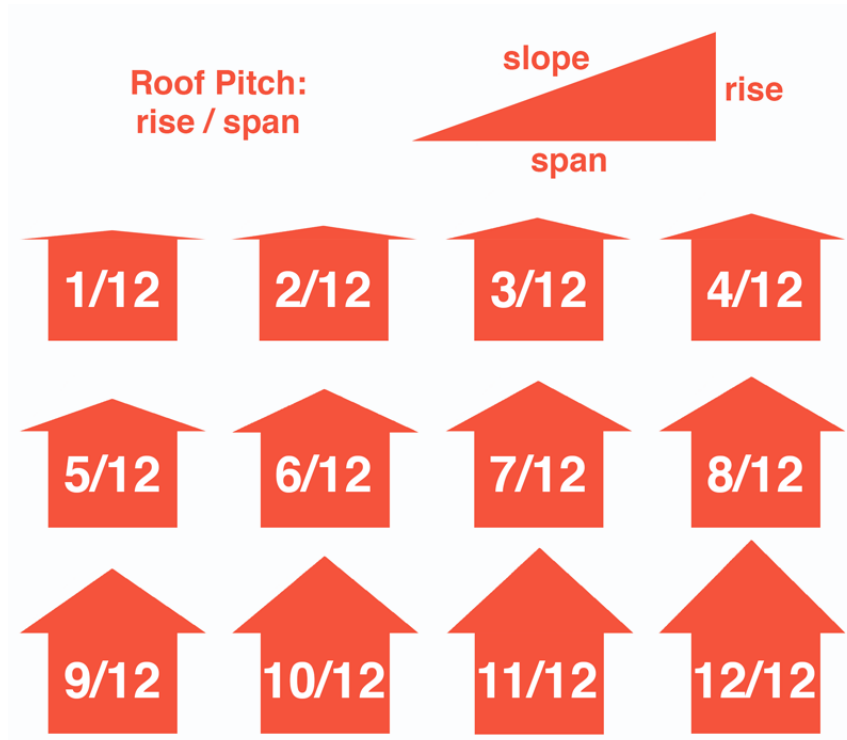
**String Inverter:** A device that is connected to a string of solar panels that collects the DC electricity created by the solar panels into AC electricity needed for the home.  
(We use Solaredge as our string inverter+optimizers )

**Title or deed:** the deed is an official written document declaring a person's legal ownership of a property, while a title refers to the concept of ownership rights.

**Land Survey:** the land survey is a drawing that shows exactly where the boundaries of a property are. A survey will also layout the dimensions and location of any buildings or other site improvements on the property.

# Key Terms

**Roof Pitch:** When building construction, roof pitch is the steepness of a roof quantified as a ratio or as the number of angular degrees that one 'exposure' surface deviates from horizontal level. A roof surface may be either 'functionally flat' or pitched.



**Azimuth:** The solar azimuth angle is the azimuth angle of the Sun's position. This horizontal coordinate defines the Sun's relative direction along the local horizon, whereas the solar zenith angle defines the Sun's apparent altitude.

